

MODEL CA6 - COPPER ANALYZER

Compact online colorimeter for the automatic measurement of Copper in water

APPLICATION FIELDS

- Wastewater
- Process water
- Industrial sewage treatment plants
- Ultrapure water treatment



ADVANTAGES / FEATURES

Dual compartment enclosure

To ensure complete separation between the electronics and the wet part.

Low reagent consumption

Minimum operating cost by small reagent consumption, only 1L (0.26 US.gal) R1, R2 and 2L (0.53 US.gal) R3 for the 16 mm cell / 2L (0.53 US.gal) R1, R2 and 4L (1.06 US.gal) R3 for the 26 mm cell of each reagent every 30 days with 15 minute analysis frequency.

Automatic calibration / validation / cleaning

Validation, cleaning and calibration are standard features which significantly reduce downtime and operator intervention ensuring the most accurate results are obtained.

Free selectable validation, cleaning and calibration intervals.

Wide measuring range

The determination ranges of the CA6 Copper Analyzer vary from trace µg/L to 150 mg/L using internal dilution module.

Color touchscreen user interface

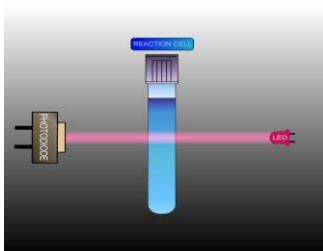
The CA6 Colorimeter is equipped with a graphic touchscreen interface showing measured values and status information. Easy access to menus and functions. Multiple languages. Integrated datalogger with USB download.

Factory tested, ready for installation and operation

Just connect the power, sample, and reagent lines and the analyzer is fully operational.

Multiple streams

Dual streams version available. External Sequencer, switching up to 4 sample streams.

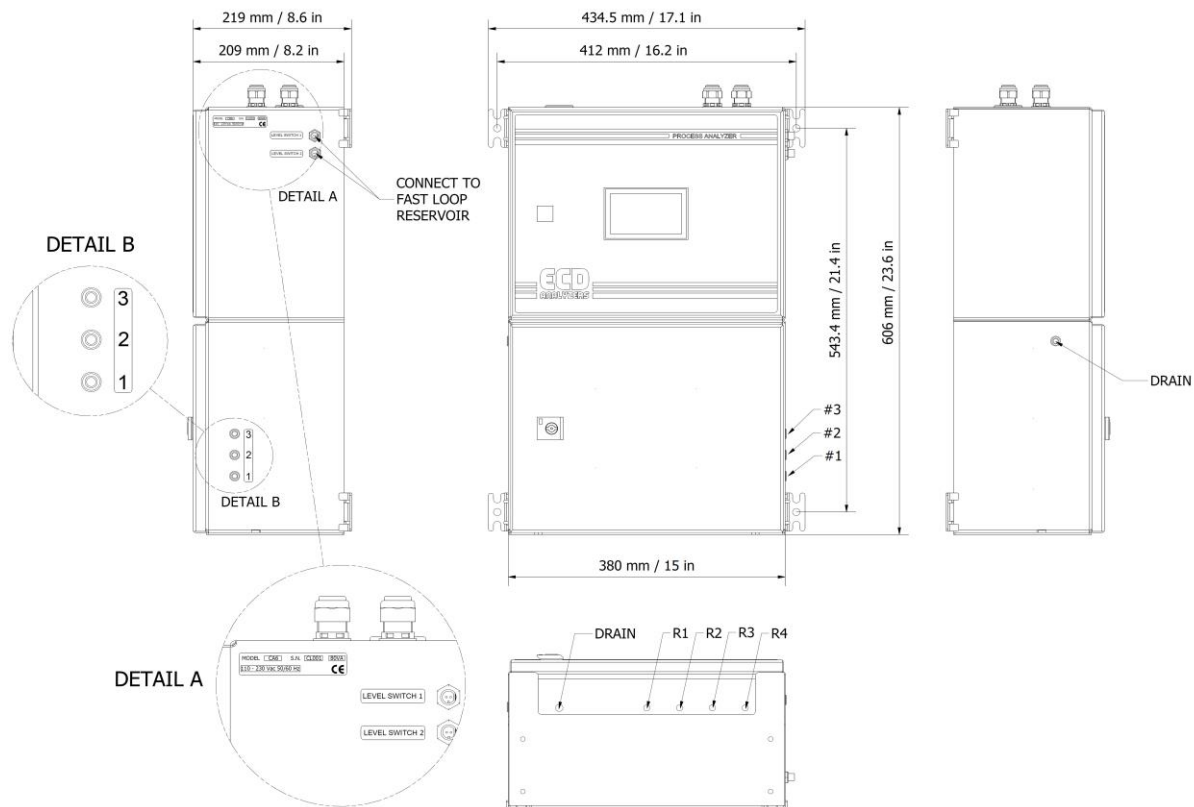


MEASUREMENT PRINCIPLE

Copper (I) ions form an orange-colored complex with the bathocuproine in a buffered solution at pH 4,5. Any copper (II) ions present in the water sample are previously reduced to copper (I) ions before the complex is formed. The absorbance intensity is proportional to the silica concentration in the sample and is determined at 470 nm.

TECHNICAL SPECIFICATIONS

Measured parameter:	Cu ⁺ / Cu ²⁺ (ppb, ppm, mg/l).	Dimensions (H x W x D):	23.6 x 15.0 x 8.2 in / 606 x 380 x 209 mm
Measuring principle:	Differential photometric absorbance. Bathocuproine Method	Weight:	Approx. 44 lbs (20 Kg)
Measuring range:	0.05 to 1 mg/L (26 mm cell) 0,1 to 3 mg/L (16 mm cell) up to 150 mg/L with internal dilution.	Power supply:	Voltage: 100 - 240 VAC 50/60 Hz standard or 24 VDC (option) Power consumption: max. 80 VA
Reproducibility:	± 20 ppb or ± 5%, whichever is greater (26 mm cell) ± 50 ppb or ± 5%, whichever is greater (16 mm cell)	Outputs:	2 x 4-20 mA outputs for measured data Modbus RTU RS485
Analysis frequency:	Freely programmable, batch near-continuous analysis.	Alarms:	4 SPDT programmable potential free relays
Cycle time:	8-10 minutes, including conditioning before analysis cycle and rinsing after measuring.	Digital input:	Remote start / stop
Reaction cell:	Temperature heated	Operating Temperature:	41 - 113 °F (5 - 45 °C)
Sample:	Pressure-free from overflow vessel Temperature: 41 - 122 °F (5 to 50 °C) Flow Rate: 80 to 500 mL/min Connection: 6 mm (¼-in.)	Humidity:	10 to 90% non-condensing (indoor use, outdoor installation only possible with protective cabinet or shelter not included)
Drain:	Pressure-free, atmospheric drain Connection: 12 mm (½-in.)	Installation:	Wall mount (standard), bench top support or panel mount (options).
N° of streams:	1, 2 with integrated switching valve 3, 4 with external sequencer	Ingress Protection:	IP54



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